Review 2 Reading Guide & Cheatsheet

Design of a Temperature-Compensated PLL Frequency Synthesiser

For Viju Sanjai — Review II, 30.09.2026

This guide explains everything done between Review 1 and Review 2, why it was done, the numbers to know, and how to answer the hard questions. Every number here comes from `results/figdata.json` or `scripts/analyze_mc.py`, the same data the slides use, plus the post-review checks in `results/vco/kvco_fine/`, `results/vco/floor_check/`, `scripts/loop_check.py` and the loop-filter retune runs in `results/loop/retune/` and `results/loop/retune_corners/`. Section 11 lists the corrections made after Review 2; read it, because the examiner may ask.

Contents

1 The story in 60 seconds **2**

2 The system: what each block does **2**
2.1 K_{VCO} per corner, and why the old number was wrong 3

3 Result 1: uncompensated drift across all corners **3**

4 Why second order: the drift is curved **4**

5 The compensation network **5**
5.1 Results at tt 5
5.2 Results at every corner 6

6 Monte Carlo and the statistics **6**

7 Closed loop and the cold-start fix **7**
7.1 Loop dynamics, honestly: found, fixed, verified 8
7.2 Lock across all five corners 9

8 Methodology and research framing **10**

9 Slide-by-slide talking points **10**

10 Numbers to memorise **12**

11 Corrections made after Review 2 **12**

12 Likely questions and answers **12**

13 Glossary **14**

1 The story in 60 seconds

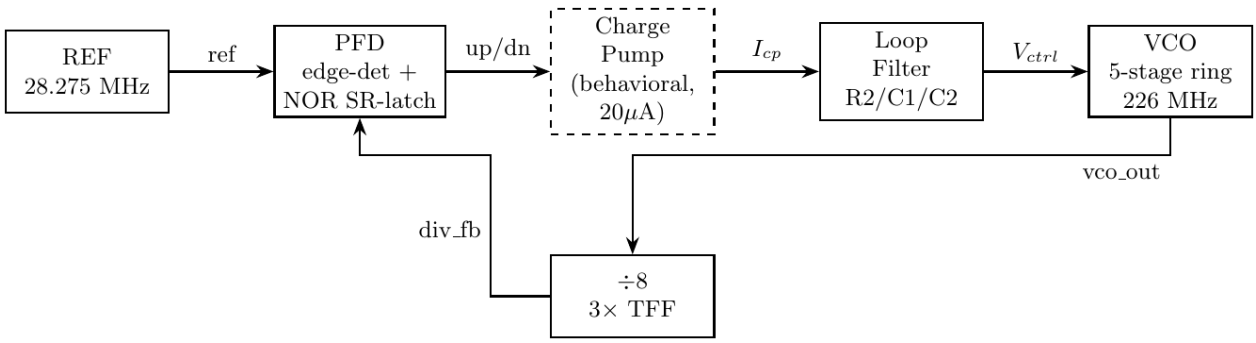
Say this

A ring-oscillator VCO changes frequency with temperature, so a PLL built around it has to work harder, or drop out of lock range, as the chip heats up. We built a complete integer-N charge-pump PLL in the open sky130 process and simulated it in ngspice. We measured the drift at all five process corners and found a 12.5× spread in temperature coefficient, not the 3.4× we reported in Review 1. We then designed a correction voltage that adds $k_1(T-27)+k_2(T-27)^2$ to the VCO control voltage. At the typical corner, the second-order version cuts the drift span from 8.51% to 1.12%. It meets our ±0.3% target only at the ff and sf corners. We validated the result at held-out temperatures and with a 30-sample Monte Carlo analysis using paired statistics. We also fixed the cold-start problem left open in Review 1: the loop now locks from a true 0 V start in 1352 ns. Auditing the loop afterwards, we found it was underdamped ($\zeta = 0.33$), retuned the loop-filter resistor from 1.7 to 3.6 kΩ, and verified by transistor-level simulation that ζ is now about 0.70 and the loop locks inside the 2 μs spec at all five corners. The price is a slower cold start (839 ns with the original filter).

The three things Review 2 added

1. **Characterisation:** the full 5-corner × 6-temperature VCO grid.
2. **Compensation:** 1st- and 2nd-order correction, fitted per corner and validated honestly.
3. **Robustness:** the cold-start fix, plus a Monte Carlo analysis with correct statistics.

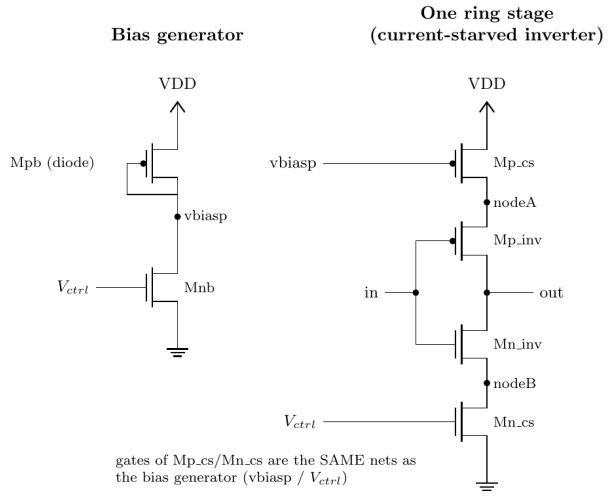
2 The system: what each block does

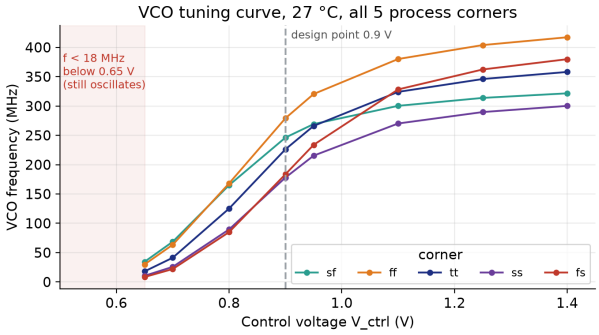


Closed-loop lock (tt, 27°C): **503 ns** (R2 = 3.6 kΩ; spec <2 μs)

- **Reference:** 28.275 MHz. The target output is 8× that, ≈226 MHz.
- **PFD (phase-frequency detector):** compares the reference edge against the divided VCO edge and outputs UP or DN pulses in proportion to the phase error. It is built from edge detectors and NOR SR-latches rather than resettable D flip-flops, because a reliable asynchronous reset fights the latch's own feedback.
- **Charge pump:** converts UP/DN pulses into current pulses of $I_{cp} = 20\,\mu$A into the loop filter. In Review 2 this is an idealised behavioural source; the transistor-level version is Review 3.
- **Loop filter:** $R_2 = 3.6\,$kΩ (retuned; the original design used 1.7 kΩ) in series with $C_1 = 68\,$pF, and $C_2 = 6.8\,$pF in parallel. It integrates the current into a control voltage V_{ctrl}. R_2 adds the zero that keeps the loop stable; C_2 smooths the ripple.
- **VCO:** a 5-stage current-starved ring. V_{ctrl} sets the starving current, which sets the delay per stage and therefore the frequency. $K_{VCO} \approx 890$ MHz/V at 0.9 V (tt, local slope). There is no hard oscillation floor: below about 0.65 V it is under 18 MHz, and at 0.5 V it is sub-MHz.

- **Divider ÷8:** a chain of toggle flip-flops. It feeds the divided clock back to the PFD.





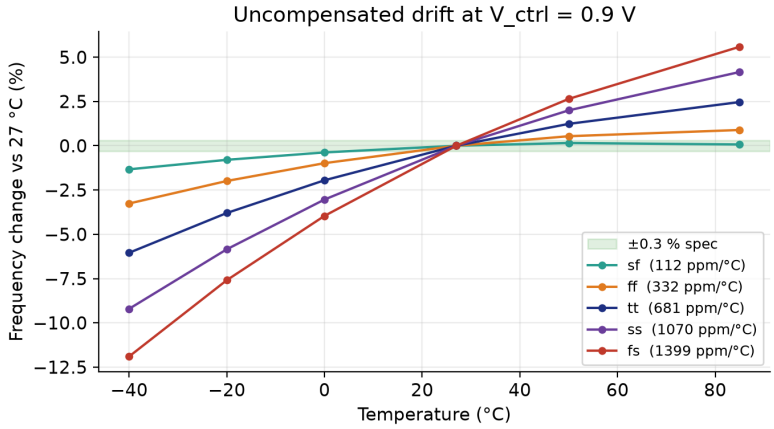
How to read the tuning curve. The slope of this curve is K_{VCO}, and it is not constant: the curve is S-shaped, 18 MHz at 0.65 V to 358 MHz at 1.40 V (tt). At 0.9 V the local slope is ≈890 MHz/V at tt. Below about 0.65 V the ring does not stop; it drops into subthreshold operation and the frequency falls by roughly 2.5–3× for every −50 mV (41.3, 18.0, 6.47, 2.09, 0.63 MHz at 0.70, 0.65, 0.60, 0.55, 0.50 V). So there is *no hard floor*, but at a 0 V start the divided clock is effectively absent, which is why the cold start was a problem.

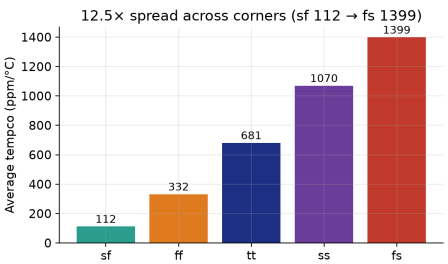
2.1 K_{VCO} per corner, and why the old number was wrong

Corner	sf	ff	tt	ss	fs
Local K_{VCO} at 0.9 V, 27 °C (MHz/V)	580	940	890	820	1020
Old Review-2 figure (MHz/V)	580	764	669	568	689

Method: central difference of f at $V_{ctrl} = 0.9 \pm 0.05$ V from a fine sweep (`results/vco/kvco_fine/`); tt gives 887 MHz/V. **Why 669 was wrong:** it came from a quadratic fit over the whole S-shaped 0.65–1.4 V curve, which averages away the steep part around 0.9 V. The sf figure happened to agree; the others were low by 1.2–1.5× (tt by 1.33×). Anything that used 669 is affected: the Review-1 loop design, and the analytic k_1 (see below). Anything that came from direct simulation (refit coefficients, spans, held-out checks, Monte Carlo, lock times) is *not* affected.

3 Result 1: uncompensated drift across all corners





- **Procedure:** fix $V_{ctrl} = 0.9$ V, sweep $T \in \{-40, -20, 0, 27, 50, 85\}$ °C, and do this at all 5 corners (30 simulations).
- **Tempco** $= (f_{85} - f_{-40})/(f_{27} \cdot 125)$ in ppm/°C. Values: sf 112, ff 332, tt 681, ss 1070, fs 1399.
- **The correction to Review 1:** with only 3 corners we reported a 3.4× spread. The full grid shows 12.5×, because fs, the worst corner, had never been simulated.
- **Insight:** the skew corners bracket the range: sf (fast-NMOS / slow-PMOS) is lowest and fs (slow-NMOS / fast-PMOS) is highest. The tempco ranks with the *NMOS drive* at 0.9 V, as the table shows.

Corner	sf	ff	tt	ss	fs
NMOS I_d at $\lvert V_{gs}\rvert = 0.9$ V, 27 °C (μA)	45.1	42.8	36.2	30.3	28.2
Tempco (ppm/°C)	112	332	681	1070	1399
f at 0.9 V, 27 °C (MHz)	246.2	279.2	226.2	178.0	183.7

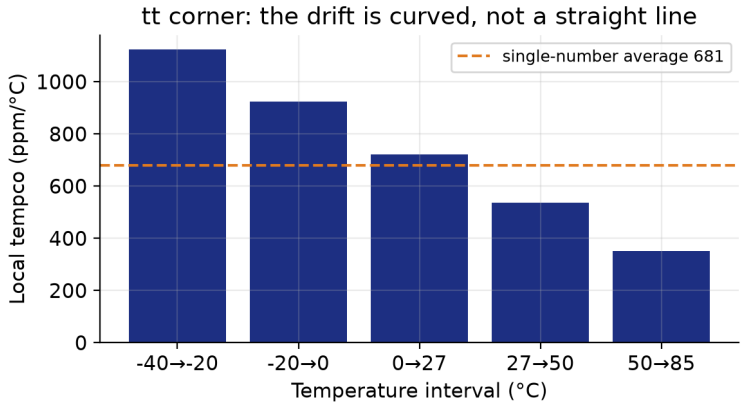
Be careful

Corner labels. In the sky130B PDK, `sf` is NMOS *fast* (threshold −27 mV, mobility +5%) with PMOS slow, and `fs` is NMOS *slow* (+27 mV, −5%) with PMOS fast. Say "sf (fast-NMOS / slow-PMOS)" and "fs (slow-NMOS / fast-PMOS)"; never "sf = slow N / fast P". The skew corners move the NMOS threshold by ±27 mV against only ±15 mV for ss/ff, so they are more extreme on the NMOS than ss/ff, which is why they bracket the range.

Say this

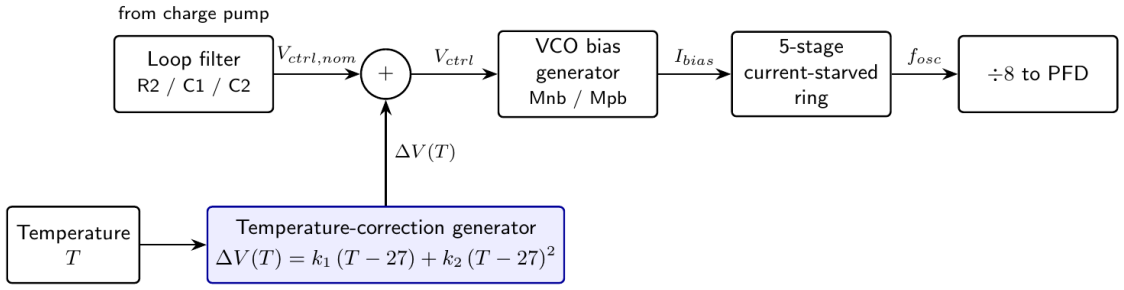
Why frequency rises with temperature (supported interpretation). V_{ctrl} drives the NMOS starving gate directly and sets the PMOS bias through the Mnb→Mpb mirror, so the NMOS overdrive sets the starving current. At 0.9 V that overdrive is small. Rising temperature lowers V_{th} and lowers mobility; with little overdrive, the V_{th} drop is a large fraction of it and beats the mobility loss, so current and frequency *rise*. Less overdrive gives a stronger positive tempco, which is exactly the ranking above (sf 112 < ff 332 < tt 681 < ss 1070 < fs 1399). Check: at tt with $V_{ctrl} = 1.4$ V (more overdrive) the tempco is *negative*, −816 ppm/°C (f = 380.2, 358.1, 343.7 MHz at −40, 27, 85 °C). So the tempco changes sign between 0.9 V and 1.4 V, i.e. a zero-tempco bias point exists. This is the textbook picture, and the sign flip plus the ordering support it. It is still an *interpretation*: we did not run a separate isolation experiment.

4 Why second order: the drift is curved



The tt tempco measured over each temperature interval falls from 1123 ppm/°C (cold) to 352 (hot). A straight line (1st order) can only match the average slope of 681 ppm/°C, so it leaves a bowed residual. Adding $k_2(T - 27)^2$ matches that bow.

5 The compensation network

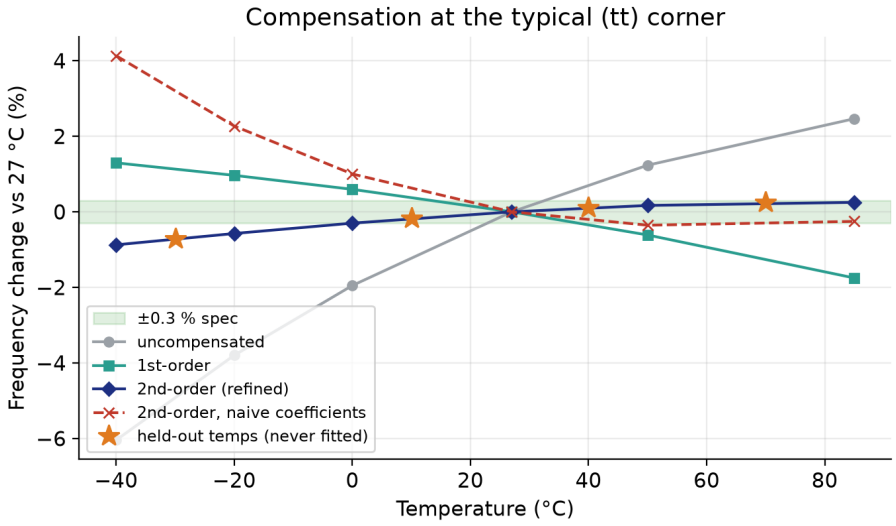


1st-order: $k_2 = 0$ (linear, PTAT/CTAT-equivalent) **2nd-order:** $k_2 \neq 0$ (curvature-matched)
Coefficients fitted per process corner; an ideal behavioural source in Review 2.

$$V_{ctrl,eff} = V_{ctrl,nom} + k_1(T-27) + k_2(T-27)^2$$

- **Implementation:** an ideal behavioural voltage source (an ngspice B-source) that reads the simulator temperature. **It is not transistors yet.** In silicon, this role is played by a PTAT/CTAT current generator.
- k_1 **(analytical):** from the slope of the fitted $f(T)$ divided by K_{VCO} (the analytic value $k_1 = -2.163 \times 10^{-4}$ V/°C used 669 MHz/V). Because the frequency rises with T, k_1 is negative.
- k_2**, first attempt (naive):** derived from the curvature of the fit. It *over-corrected* the cold end (+4.13% at −40 °C, and a 4.49% span, worse than 1st order's 3.05%). There were two causes: it assumed K_{VCO} is constant, which it is not, and it used the wrong K_{VCO} (669 instead of the true local ≈887 MHz/V), which overestimated the needed correction by about 1.3×.
- k_2**, refined:** refit against the *measured* residual left after 1st-order compensation. This step needed no extra simulation. At tt: $k_1 = -1.320 \times 10^{-4}$ V/°C and $k_2 = 2.891 \times 10^{-7}$ V/°C^2.
- **Per corner:** k_1 and k_2 are refit separately at every corner. In silicon this corresponds to a per-corner trim.

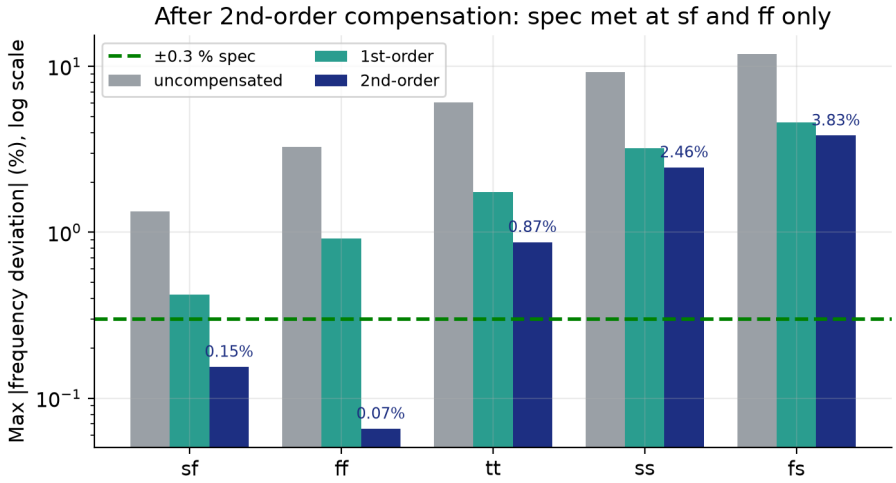
5.1 Results at tt



- The span goes from 8.51% (uncompensated) to 3.05% (1st order) to 1.12% (2nd order), so $C = 8.51/1.12 = 7.57\times$.
- The maximum |deviation| after 2nd order is 0.87%, which does **not** meet ±0.3%.

- **Held-out check (the orange points):** −30, 10, 40 and 70 °C were never used in fitting, and all of them fall inside the fitted envelope. This is our guard against overfitting 6 points with a 2-coefficient curve.

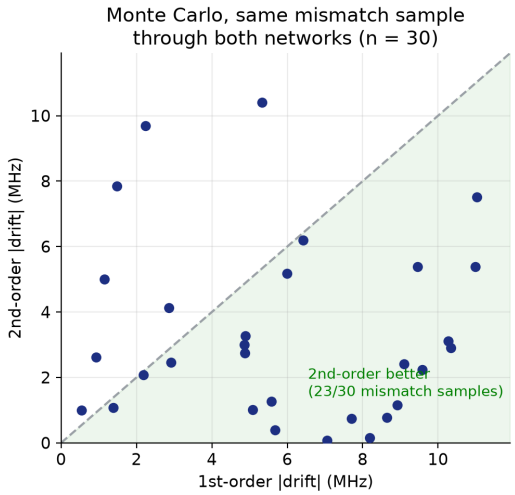
5.2 Results at every corner

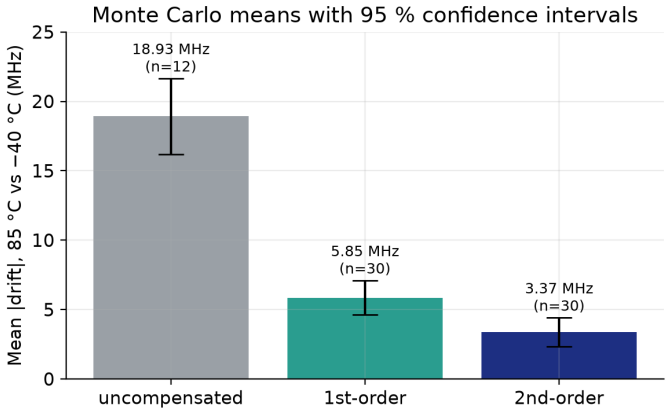


Corner	Baseline	1st order	2nd order	C (2nd)	max|dev|	Meets ±0.3%?
ff	4.15%	1.06%	0.10%	40.3×	0.07%	yes
sf	1.48%	0.42%	0.24%	6.1×	0.15%	yes
tt	8.51%	3.05%	1.12%	7.6×	0.87%	no
ss	13.37%	6.37%	3.32%	4.0×	2.46%	no
fs	17.49%	9.13%	5.31%	3.3×	3.83%	no

Every corner improves at both stages. **Span** means max − min frequency over −40 to 85 °C, as a percentage of f_{27}. C ranges over 12× across corners, which is why the project charter defines C per corner and not as one number.

6 Monte Carlo and the statistics





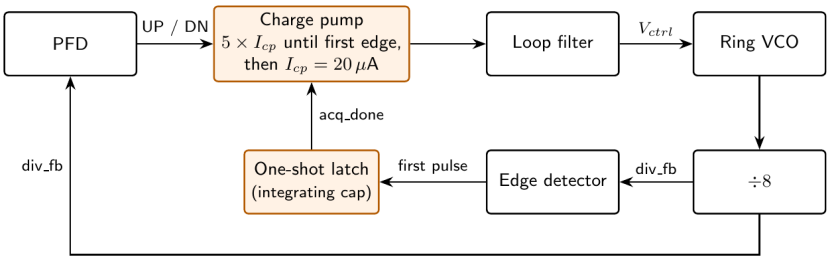
- **Setup:** tt corner with sky130's mismatch models switched on (`mc_mm_switch=1`). Each *seed* is one virtual chip with random transistor mismatch.
- **Metric:** |drift| = $|f_{85} - f_{-40}|$ in MHz.

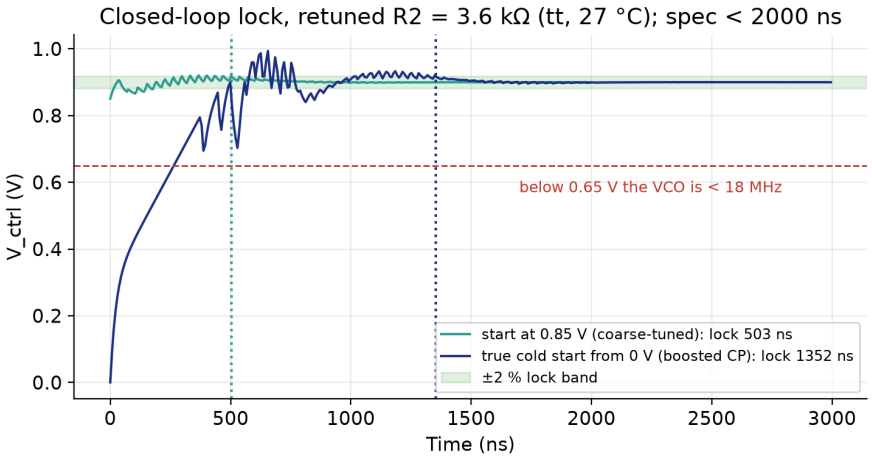
- **Paired design:** the *same* seed runs through both the 1st- and 2nd-order networks, so chip-to-chip variation cancels out of the comparison.
- **Results:** mean |drift| is 18.93 MHz uncompensated (n=12), 5.85 MHz for 1st order (n=30) and 3.37 MHz for 2nd order (n=30).
- **2nd vs 1st:** a reduction of 2.48 MHz, 95% CI [0.81, 4.15]. The paired t-test and the Wilcoxon signed-rank test are both significant, and 2nd order wins in 23 of 30 seeds.
- **Two tests:** the t-test assumes normality; Wilcoxon does not. Both agreeing makes the result robust. Both are hand-implemented in `scripts/stats.py` (there is no scipy in this environment) and self-tested.

Be careful

The correction we made ourselves (own it, it is a strength). The first analysis compared *signed* drift. First order over-corrects (its drift is negative, −5.7 MHz) while second order under-corrects slightly (+2.3 MHz). Comparing signed values therefore made the difference look large simply because the sign flipped. The right quantity is |drift|. On |drift| at n=12 the difference was *not* significant, so we raised n to 30, and that is what earns the claim. This is recorded in `docs/stats_methodology.md` and in the project charter's decision log.

7 Closed loop and the cold-start fix



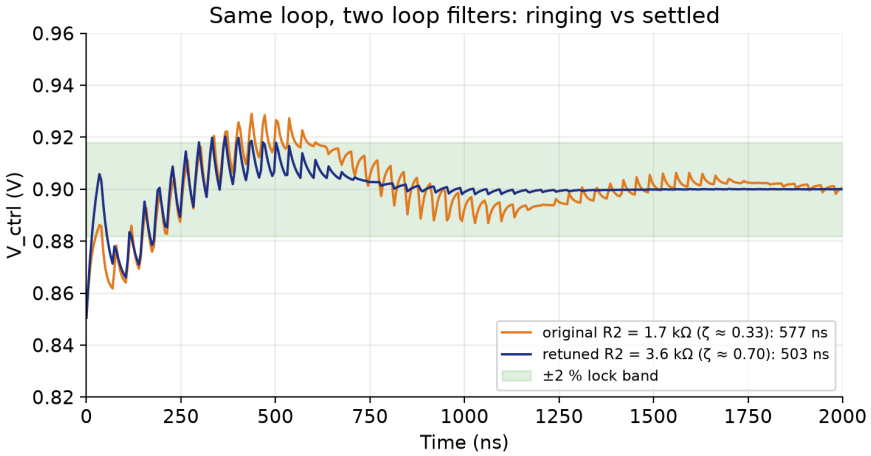


- **The problem:** at 0 V the VCO is essentially off. There is no hard floor, but below about 0.65 V it runs under 18 MHz (under 8% of the 226 MHz target) and at 0.5 V below 1 MHz, so the divided clock is effectively absent. The PFD sees only reference edges, and pumping up at the nominal 20 μA is too slow.
- **The fix:** the charge pump runs at $5 \times I_{cp}$ until the divider produces its first edge. An edge detector feeding a one-shot latch (an integrating capacitor charged by a B-source, not a digital latch with an unknown initial state) then drops it back to 20 μA for normal, stable tracking.

- **Result with the retuned loop filter ($R_2 = 3.6$ kΩ):** lock in **1352 ns** from a true 0 V start, and **503 ns** from a coarse-tuned 0.85 V start (tt, 27 °C). Both are inside the <2 μs spec.
- **Original design ($R_2 = 1.7$ kΩ), kept as the "before":** 839 ns from 0 V and 577 ns from 0.85 V. (Review 1's "450 ns analytical prediction" matched only by coincidence; see the loop-dynamics subsection.)
- **Lock definition:** V_{ctrl} enters and stays within ±2% of its final 0.9 V.
- **The cold start is slower after the retune (839 → 1352 ns, +61%).** We do not hide this: it is the price of the better damping. Cold-start peak V_{ctrl} is 1.001 V (1.008 V originally). Do not claim that the retune made everything faster; only the coarse-start lock is faster.
- **The ringing around 600–800 ns on the original cold-start curve** is the loop capturing phase after the fast pump-up. It settled inside the spec, but the original loop was genuinely underdamped (next subsection). We have not separately attributed it to the boost switching off, so don't claim that.

tt, 27 °C	Original, $R_2 = 1.7$ kΩ	Retuned, $R_2 = 3.6$ kΩ
Lock, coarse 0.85 V start	577 ns	503 ns
Lock, true 0 V cold start	839 ns	1352 ns
Peak V_{ctrl} (coarse start)	0.930 V	0.922 V
Final f_{VCO} (coarse start)	227.96 MHz (0.78% high)	226.18 MHz (0.008% low)
V_{ctrl} ripple, last 0.5 μs	8.6 mV p-p	0.35 mV p-p

An intermediate $R_2 = 3.0$ kΩ was also simulated: 538 ns (coarse) and 1318 ns (cold start).



Coarse 0.85 V start, tt: original (orange) rings and is still about 0.8% off in frequency at 2 µs; retuned (blue) settles smoothly.

7.1 Loop dynamics, honestly: found, fixed, verified

- **Found:** with the original $R_2 = 1.7$ kΩ, $C_1 = 68$ pF, $I_{cp} = 20\,\mu$A, $N = 8$ and the measured $K_{VCO} = 887$ MHz/V, the natural frequency is $f_n \approx 0.91$ MHz and the damping is $\zeta \approx 0.33$. Phase margin is about 30–33° (the 100ζ rule; C_2 erodes it a little more). The loop was **underdamped**: from a 0.85 V start V_{ctrl} overshot to 0.93 V, and the cold start rang.
- **What Review 1 got wrong:** it put K_{VCO} in rad/s/V into $\omega_n^2 = I_{cp}K_{VCO}/(NC_1)$ and *dropped the* $1/2\pi$ of the PFD/charge-pump gain ($I_{cp}/2\pi$ per radian). The correct form is $\omega_n^2 = I_{cp}K_{VCO,rad}/(2\pi NC_1)$ and $\zeta = R_2C_1\omega_n/2$. Review 1 therefore claimed $f_n = 1.98$ MHz, $\zeta = 0.72$, 70–72° margin, 2.85 MHz bandwidth and a 450 ns lock; with the same wrong 669 MHz/V and the right formula one gets $f_n = 0.79$ MHz and $\zeta = 0.29$.

- **Fixed:** ζ depends only on R_2 at fixed f_n, so we raised R_2 from 1.7 to **3.6 kΩ**. f_n stays about 0.91 MHz, ζ goes from 0.33 to about **0.70**, and the phase margin from about 30° to about **65–70°** (100ζ rule). Nothing else in the loop changed (I_{cp}, C_1, C_2, N, f_{ref}).
- **Verified at transistor level (tt, coarse start):** ripple 8.6 → 0.35 mV p-p, final frequency error 0.78% → 0.008%, peak V_{ctrl} 0.930 → 0.922 V, lock 577 → 503 ns. The independent averaged model (`scripts/loop_check.py`, driven by the measured tuning curve) gave 573 ns against 577 ns for the original loop, so it was a trustworthy check of the measured lock times; the 450 ns "prediction" matched only by coincidence. Without the 5× boost, the averaged model from 0 V does not settle (the pump saturates, overshoot 1.79 V), consistent with the cold-start problem.
- **The trade-off:** the 0 V cold start is slower (839 → 1352 ns). Both are within the 2 μs spec.

Be careful

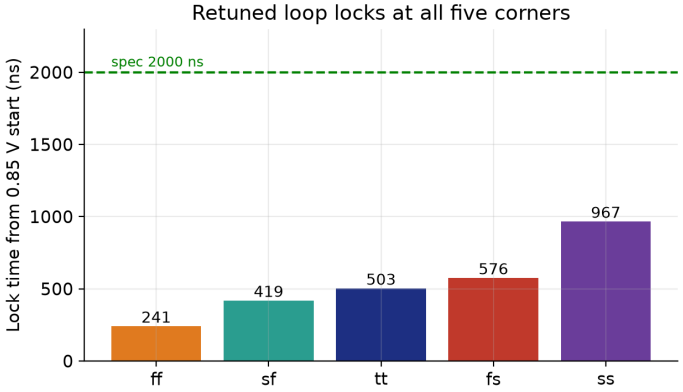
Limits of the retune claim. Only $R_2 = 1.7$, 3.0 and 3.6 kΩ were simulated at transistor level (plus an averaged-model sweep of 1.7–6 kΩ). We do not claim 3.6 kΩ is optimal, and values above 3.6 kΩ were not simulated at transistor level. The 503 ns figure is set by a small overshoot to 0.922 V that sits just outside the ±2% band (0.918 V) and decays into it; the averaged model has a slightly smaller overshoot (0.914 V) and enters the band at about 110 ns. The models agree on the dynamics, but the ±2% band metric is sensitive at the band edge, so quote 503 ns as a transistor-level number, not as a precise property of the loop. The 0 V cold start was simulated at tt only.

7.2 Lock across all five corners

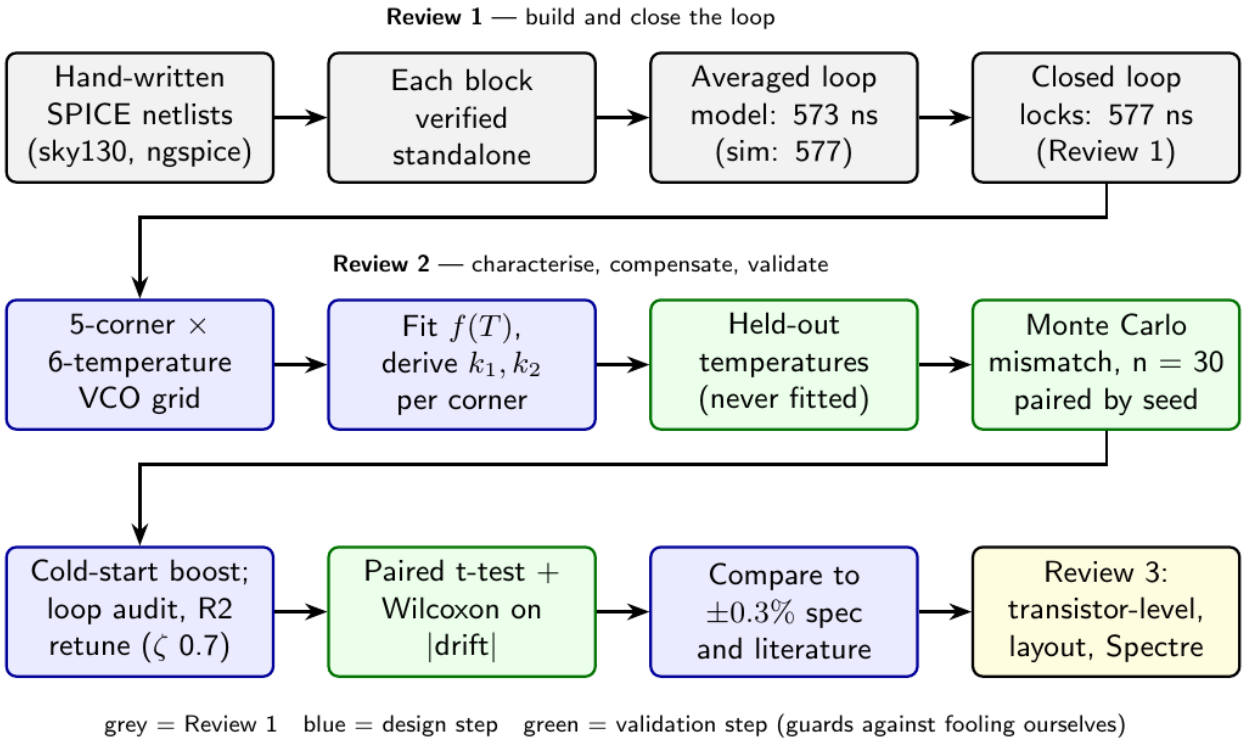
Coarse 0.85 V start, nominal charge pump (no boost), as in the tt coarse run. Lock is V_{ctrl} staying within ±2% of its final value; spec <2 μs.

Corner	Retuned lock	Retuned final f_{VCO} (MHz)	Original ($R_2 = 1.7$ kΩ)
tt	503 ns	226.18	577 ns, 227.96 MHz
ff	241 ns	226.17	not settled by 2 μs: 225.17 MHz (0.45% low)
sf	419 ns	226.39	not settled by 2 μs: 229.13 MHz (1.3% high)
fs	576 ns	226.21	not settled by 2 μs: 228.90 MHz (1.2% high)
ss	967 ns	224.56 (2 μs run, still settling)	1737 ns in a 5 μs run; at 2 μs it was 3.6% low (218.2 MHz)

- The retuned loop locks inside the 2 μs spec at all five corners. It is slowest at ss (967 ns; ss needs $V_{ctrl} \approx 0.968$ V, the corner with the weakest NMOS). The ss lock time and its settled value ($V_{ctrl} = 0.9677$ V, 0.1 mV ripple) come from a 5 μs run; in the 2 μs run ss is still settling at 224.56 MHz.
- **Honest note:** the original loop had *not* finished settling at ff, sf, fs or ss within its 2 μs run. So we quote no original lock time for those corners, except ss (from the 5 μs run, 1737 ns, 3.5 mV ripple). The comparison there is the frequency error left at 2 μs, not a time.



8 Methodology and research framing



- **Hypothesis H1 (the research question):** the error in a compensation network's residual grows as $\varepsilon_{res} \approx C \cdot \varepsilon_{raw}$. In words, the better you compensate, the more a small error in the drift *measurement* (for example, from the open PDK's models versus a commercial one) is magnified in what remains. Testing it needs a second tool flow (Cadence Virtuoso/Spectre), which is **pending lab access**. The strength (α) sweep we ran provides a range of C values ready for that test; on its own it is *not* the H1 test.
- **Novelty (be precise):** US Patent 9,748,962 B2 (Infineon, 2017) already covers generic quadratic temperature compensation. We do *not* claim quadratic compensation as new. Our claim is narrower: a quantified, statistically validated, per-corner study of it for a *ring-VCO PLL, closed-loop, on an open PDK*.

9 Slide-by-slide talking points

Numbers follow the 27-slide deck: 25 is Challenges, 26 References, 27 Thank you.

1. **Title.** Names, guide, date.
2. **Introduction.** Problem, objectives, and what changed since Review 1: the 5-corner grid, the cold-start fix, and 2nd-order compensation.

3–4. **Literature.** 15 IEEE papers. The gap: nobody combines a compensated ring VCO, a closed-loop PLL, an open PDK and statistical validation.

5. **Architecture.** Walk the loop from REF to ÷8, and name the two deliberate simplifications (ideal charge pump, latch-based PFD).
6. **VCO.** ≈890 MHz/V local slope at 0.9 V (tt); no hard floor, under 18 MHz below 0.65 V; 0.9 V gives 226 MHz.
7. **Compensation architecture.** $\Delta V(T)$ added before the bias generator; 1st vs 2nd order; per corner.

8. **Cold-start fix.** Problem, then the 5× boost until the first edge. Lock from 0 V is now 1352 ns with the retuned loop filter.
9. **Methodology.** Grey is Review 1, blue is design, green is validation. The green steps are there to stop us fooling ourselves.
10. **Implementation.** The modules, the coefficient derivation, the hand-written statistics, and the B-source line.
11. **Results table.** 12.5× spread; tempco ranks with NMOS drive (sf fast-NMOS lowest, fs slow-NMOS highest).
12. **Drift chart.** Every corner is outside ±0.3%; fs is the worst.
13. **Why the corners order this way.** Tempco versus NMOS drive current: sf 45 μA (112 ppm/°C) to fs 28 μA (1399). Say sf = fast-NMOS, fs = slow-NMOS, and that the tempco flips sign at 1.4 V (−816 ppm/°C).
14. **Curvature.** The bars would be equal if the drift were linear; they fall 3.2×, which justifies k_2.
15. **Per-corner table.** Everything improves; the spec is met at ff and sf only.
16. **tt compensation chart.** 8.51 → 1.12%; the naive curve (red dashed) failed; the held-out points pass.
17. **Per-corner chart.** Log scale; the green line is the spec. Say plainly: "not met at tt, ss or fs".
18. **Closed-loop table (both designs).** Original R_2 = 1.7 kΩ against retuned 3.6 kΩ: coarse lock 577 → 503 ns, cold start 839 → 1352 ns, ζ 0.33 → 0.70, ripple 8.6 → 0.35 mV. State the trade-off yourself: the cold start got slower.
19. **Lock transient (retuned loop).** 503 ns from 0.85 V, 1352 ns from a true 0 V start (it was 839 ns with the original filter: say so). Point at the boost carrying V_{ctrl} quickly through the region below ∼0.65 V where the VCO is nearly off (not a hard floor) on the cold-start curve.
20. **Loop dynamics.** The chart overlays original (orange: rings, still off at 2 μs) and retuned (blue: settles). Found (Review 1 dropped $1/2\pi$, ζ was 0.33), fixed (R_2 3.6 kΩ, $\zeta \approx 0.70$, PM ∼65–70°), verified (transistor-level). Say that only 1.7 / 3.0 / 3.6 kΩ were simulated and 3.6 is not claimed optimal.
21. **Lock across all five corners.** ff 241, sf 419, tt 503, fs 576, ss 967 ns, all inside 2 μs. Say plainly that the original loop had not settled at ff/sf/fs/ss within its 2 μs run, so there is no old lock time to compare at those corners.
22. **MC and benchmark table.** Comparable to published work, but ours is simulation and theirs is silicon.
23. **MC charts.** Points below the diagonal are chips where 2nd order won (23/30).
24. **At a glance.** Eight numbers.
25. **Challenges and remaining work.** The honest list, then Review 3.

10 Numbers to memorise

Reference / output / N	28.275 MHz / ≈226 MHz / 8
Loop filter, I_{cp}	R_2 3.6 kΩ (retuned; original 1.7), C_1 68 pF, C_2 6.8 pF; 20 μA
K_{VCO} (tt, 0.9 V, local)	≈890 MHz/V (887); sf/ff/ss/fs 580/940/820/1020
VCO at low V_{ctrl}	no hard floor; 18.0 MHz at 0.65 V, 0.63 MHz at 0.5 V
Loop: f_n, ζ, PM	0.91 MHz; ζ 0.33 → 0.70; PM ∼30 → ∼65–70° (retune)
Loop ripple, final error (tt)	8.6 → 0.35 mV; 0.78% → 0.008%
Tempco sf/ff/tt/ss/fs	112 / 332 / 681 / 1070 / 1399 ppm/°C (12.5×)
tt span: base → 1st → 2nd	8.51 → 3.05 → 1.12%, C = 7.57×
tt max|dev| (2nd)	0.87% (spec ±0.3%)
Spec met at	ff (0.07%), sf (0.15%)
MC mean |drift|	18.93 / 5.85 / 3.37 MHz
2nd vs 1st	−2.48 MHz, CI [0.81, 4.15], 23/30 seeds
Lock, tt, retuned (original)	503 ns (577) from 0.85 V; 1352 ns (839) from 0 V; spec <2 μs
Lock, retuned, 0.85 V start	ff 241, sf 419, tt 503, fs 576, ss 967 ns
Temperatures	−40, −20, 0, 27, 50, 85; held-out −30, 10, 40, 70

11 Corrections made after Review 2

Corrections made after Review 2 (own these)

After the slides were made we re-checked the work and corrected five things and fixed one of them; say so openly. **(1)** The sf/fs labels were backwards: sf is fast-NMOS / slow-PMOS and fs is slow-NMOS / fast-PMOS. **(2)** The tempco explanation ("mismatched corners balance the current sources") was wrong; the tempco follows NMOS overdrive, and the sign flips at higher V_{ctrl} (tt, 1.4 V: −816 ppm/°C). **(3)** K_{VCO} was 669 MHz/V from a quadratic fit over the whole curve; the local value at 0.9 V is ≈890 MHz/V (887), which also explains part of the analytic k_2 over-correction. **(4)** Review 1's loop design dropped a $1/2\pi$, so the original loop ($R_2 = 1.7$ kΩ) was underdamped ($\zeta \approx 0.33$, ∼30° margin, not 0.72 and 70°), though its measured lock times were confirmed by an independent model (573 vs 577 ns). *This is now fixed and verified:* R_2 was retuned to 3.6 kΩ ($\zeta \approx 0.70$, PM ∼65–70°) and re-simulated at transistor level: ripple 8.6 → 0.35 mV, frequency error 0.78% → 0.008%, coarse lock 577 → 503 ns, cold-start lock 839 → 1352 ns (slower, within spec), and lock inside 2 μs at all five corners (ss slowest, 967 ns). **(5)** The "0.63 V oscillation floor" was an artefact of a short simulation window; the ring keeps oscillating at low frequency. **Unchanged:** spans, tempcos (112/332/681/1070/1399), the 2nd-order results, held-out checks, Monte Carlo statistics and the open-loop results are direct simulation results; the compensation and Monte Carlo use the free-running VCO with an ideal control voltage, so the loop retune does not touch them.

12 Likely questions and answers

Q: Your spec is ±0.3% and you only meet it at 2 of 5 corners. Isn't that a failure?
It is a gap, and we say so on the slide. The correction is currently a fixed curve per corner. Closing the gap at tt/ss/fs needs a transistor-level network whose PTAT slope itself tracks process, and/or a trim. That is Review 3 work. What Review 2 establishes is how large the problem is, and that 2nd order beats 1st order with statistical confidence.

Q: The compensation is an ideal source. How real is this?
It is a behavioural model, as is standard at this stage: it tells us *what* correction voltage is needed. In silicon it would be a PTAT/CTAT current generator driving a resistor; a quadratic term can come from combining PTAT and a squared term (for example, a translinear stage). Building that at transistor level is Review 3.

Q: Why a ring VCO and not an LC oscillator?

A ring is small, has a wide tuning range and needs no inductor, which suits a digital-friendly open PDK. Its disadvantage is exactly what we study: large temperature sensitivity (and worse phase noise).

Q: Why is there no phase noise or jitter result?

Out of scope for Review 2, which focuses on temperature. A transient-noise or PSS-style analysis in ngspice is limited; this would be part of the Spectre arm.

Q: Why sky130 and ngspice?

They are fully open, so the work is reproducible by anyone; the PDK is pinned by commit via volare. The trade-off is less mature models, which is exactly why H1 plans a Spectre cross-check.

Q: How do you know you didn't overfit? Six points, two coefficients.

Four held-out temperatures that were never used in fitting, at every corner, all fall inside the fitted envelope. The Monte Carlo analysis also tests the same coefficients on 30 mismatched chips.

Q: Why was the naive k_2 wrong?

It assumed K_{VCO} is constant with temperature and voltage. It is not: the tuning curve is nonlinear. It also used 669 MHz/V instead of the true local ≈887, which overestimated the correction about 1.3×. Refitting against the measured 1st-order residual absorbs both.

Q: Why is Monte Carlo only at tt, and why is the baseline n=12?

Compute time. The baseline effect is huge (12/12 seeds, reduction CI [9.98, 15.57] MHz), so n=12 is already decisive there. The 1st-vs-2nd difference is small, so it needed n=30.

Q: What was the statistics error?

We compared signed drift, but 1st order over-corrects and 2nd order under-corrects slightly, so the sign flip inflated the difference. We switched to |drift|, found it not significant at n=12, and re-ran at n=30, where it is significant. It is documented in the repo.

Q: What is new, given the Infineon patent?

Quadratic compensation itself is not new. What is new is the quantified, per-corner, statistically validated study on a ring-VCO PLL in closed loop on an open PDK, and the H1 error-amplification question.

Q: Why does the frequency increase with temperature?

At 0.9 V the NMOS starving transistor has little overdrive, so the fall of V_{th} with temperature beats the fall of mobility and the current rises. At higher V_{ctrl} the tempco flips sign: tt at 1.4 V is −816 ppm/°C, so a zero-tempco bias point exists. The sign flip and the corner ordering support this; it is still an interpretation, because we did not run a separate isolation experiment.

Q: Is your loop stable? What is your phase margin?

Yes, and now well damped. With the measured $K_{VCO} = 887$ MHz/V and the retuned $R_2 = 3.6$ kΩ: $f_n \approx 0.91$ MHz, $\zeta \approx 0.70$, phase margin about 65–70° (the 100ζ rule, an estimate, not a measured Bode plot). The original $R_2 = 1.7$ kΩ gave $\zeta \approx 0.33$ and about 30°: stable but underdamped. Review 1 claimed 0.72 and 70° for that filter only because it dropped the $1/2\pi$ in the charge-pump gain. We found that, retuned, and verified it by transistor-level simulation.

Q: Why did you retune R_2, and what did it cost?

Because the corrected formulas ($\omega_n^2 = I_{cp}K_{VCO,rad}/(2\pi NC_1)$, $\zeta = R_2C_1\omega_n/2$) showed the original 1.7 kΩ gave $\zeta = 0.33$. Raising it to 3.6 kΩ keeps $f_n \approx 0.91$ MHz and gives $\zeta \approx 0.70$. Verified at transistor level (tt, 0.85 V start): ripple 8.6 → 0.35 mV p-p, final frequency error 0.78% → 0.008%, lock 577 → 503 ns. The cost: the true 0 V cold start is slower, 839 → 1352 ns (+61%). Both are inside the 2 μs spec. We simulated only 1.7, 3.0 and 3.6 kΩ, so we do not claim 3.6 is optimal.

Q: Does it lock at every corner?

Yes, the retuned loop does, from a 0.85 V start with the nominal charge pump: ff 241 ns, sf 419, tt 503, fs 576 and ss 967 ns, all inside 2 μs. ss is slowest because it needs $V_{ctrl} \approx 0.968$ V and has the weakest NMOS. The ss figure is from a 5 μs run (in the 2 μs run it is still settling at 224.56 MHz). Honest caveat: the original loop had not finished settling at ff/sf/fs/ss within its 2 μs run (0.45% low, 1.3% high, 1.2% high and 3.6% low at 2 μs), so we give no old lock times there, except ss at 1737 ns from the 5 μs run. The 0 V cold start was simulated only at tt.

Q: Why does sf have a lower tempco than ff? Isn't ff the fastest?

sf is fast-NMOS / slow-PMOS, and the NMOS drive sets the starving current. sf has the most NMOS drive (45.1 μA at 0.9 V against 42.8 for ff), so the threshold-versus-mobility trade is closest to its zero-tempco point. The skew corners shift NMOS V_{th} by $\pm$27 mV against $\pm$15 mV for ss/ff, so they are the extremes. Note ff is the fastest in frequency (279 MHz); sf is only the fastest in NMOS current.

Q: How did you get K_{VCO}?

As the local slope at 0.9 V: a central difference at $\pm$50 mV from a fine sweep, 887 MHz/V at tt (about 890). The earlier 669 came from a quadratic fit across the whole S-shaped 0.65–1.4 V curve and was wrong; we found and corrected it. Per corner it is 580 / 940 / 890 / 820 / 1020 for sf / ff / tt / ss / fs.

Q: Does the VCO really stop at 0.63 V?

No. That was an artefact of a short simulation window. With long windows the ring keeps oscillating in subthreshold: 41.3, 18.0, 6.47, 2.09 and 0.63 MHz at 0.70, 0.65, 0.60, 0.55 and 0.50 V. The correct statement is that there is no hard floor, but below about 0.65 V the VCO is under 8% of the target, so from a 0 V start the divided clock is effectively absent. The cold-start fix is unchanged: the PFD sees only reference edges and the nominal 20 μA pump is too slow.

Q: What does the acquisition boost do to stability?

It is only active before the first divider edge, while the loop is still open. After that, the charge pump runs at the nominal 20 μA for which the loop was designed, (and for which $\zeta \approx 0.70$ with the retuned R_2), so stability during tracking is unaffected by the boost. The retune does change the cold start itself: from 0 V it now takes 1352 ns instead of 839 ns.

Q: Why a latch-based PFD?

A resettable-D-flip-flop PFD needs a clean asynchronous reset. In our hand-built cells, that reset fought the latch's bistable feedback. Edge detectors plus NOR SR-latches were reliable, and the PFD was verified standalone in Review 1.

Q: What is left for Review 3?

A transistor-level charge pump and compensation network; layout in Magic with DRC and LVS (netgen); post-layout simulation; and the Spectre arm with the H1 regression. The deadline is 28.10.2026.

13 Glossary

Process corner

The five sets of device models: tt (typical), ff/ss (both fast/slow), sf/fs (sf: NMOS fast + PMOS slow; fs: NMOS slow + PMOS fast).

Tempco

The fractional frequency change per °C, in ppm/°C.

Span

Max − min frequency over temperature, as a percentage of the 27 °C frequency.

Compensation ratio C

Baseline span divided by compensated span. Bigger is better.

PTAT / CTAT

Proportional / complementary to absolute temperature: bias sources that rise or fall linearly with T.

K_{VCO}

The VCO gain, df/dV_{ctrl}, quoted as the local slope at 0.9 V (about 890 MHz/V at tt).

Damping ratio ζ / phase margin

How well damped the loop is; $\zeta = R_2 C_1 \omega_n / 2$. Phase margin is about 100ζ degrees. Original loop: 0.33, ∼30°; retuned: 0.70, ∼65–70°.

Lock time

Time until V_{ctrl} stays within $\pm$2% of its final value.

Seed
One Monte Carlo sample, i.e. one random mismatch instance (a virtual chip).

Paired t-test / Wilcoxon
Tests on per-seed differences. The t-test assumes normality; Wilcoxon uses ranks and does not.

95% CI
The range that contains the true mean difference with 95% confidence. If it excludes 0, the effect is significant.

Held-out
Data points kept out of the fit and used only to test it.